

Appendix R8: Non-Linear Models

Interaction, Polynomial, Piecewise and Spline Models

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2024-06-22

This Quarto document contains R code examples about non-linear predictive models. It covers binary x continuous (B x C) interaction models, polynomial regressions, step function models, piecewise linear spline regression and multivariate adaptive regression splines (MARS). These models are necessary when the OLS assumption of linearity does not hold, i.e., the relationship between one or more predictors and the outcome variable is/are not linear. For the corresponding conceptual material associated with these appendices, consult the main book [Predictive Analytics and Machine Learning for Managers](#).

Table of contents

Non-Linearity	1
Binary x Continuous (B x C) Interaction Models	2
Overview	2
B x C Interactions	2
Polynomial Regression	6
Predictions and CV Testing with Polynomials	10
Step Functions	11
Piecewise Linear Spline Regression	12
Multivariate Adaptive Regression Spline (MARS)	16
Polynomial and Cubic (Basis) B-Splines	16
Polynomial and Cubic (Natural) N-Splines	17

This script was created by J. Alberto Espinosa for educational and training purposes. Feel free to use this material for your own work, but please do not share or duplicate without the author's permission.

```
RNGkind(sample.kind = "default") # To use the R default RNG
```

Non-Linearity

This appendix provides a number of modeling alternatives to use when the OLS assumption of linearity does not hold. This can be evaluated visually with plots and also tested with ANOVA or CV tests to evaluate whether some non-linear models have more predictive accuracy than the corresponding linear model. In this section, I discuss **interaction, polynomial, piecewise splines and MARS (spline)** models.

Binary x Continuous (B x C) Interaction Models

Overview

An interaction model contains at least 2 predictors and a multiplicative term between the two predictors. The regression coefficients for the 2 predictors are called **main effects** and the regression coefficient for the **multiplicative** term is called the **interaction effect**. These models are necessary when business intuition suggests that the effect of one predictor is influenced by the presence of the other predictor. Interaction effects cannot be interpreted by themselves, but need to be interpreted jointly in conjunction with their respective main effects. The interpretation of effects in an interaction model can be done from the perspective of either of the main predictors. For example, if you have an interaction model like $y \sim x_1 + x_2 + x_1 * x_2$, the total effect of x_1 will be equal to the **main effect** of x_1 plus the **interaction effect** of $x_1 * x_2$. That is, the effect of x_1 will be influenced by the presence of x_2 . If the sign of the main effect of x_1 is the same as the sign of the interaction term, the interaction effect of x_2 on x_1 is **positive** because the interaction effect **enhances** the effect of x_1 , regardless of whether these effects are positive or negative. The fact that the main and interaction effects have the same signs means that the main effect is enhanced by the interaction effect. Conversely, if the sign of the main effect of x_1 and the interaction effect have opposite signs, then the interaction is **negative** because it **offsets** the main effect. The fact that the main and interaction effects have opposite signs means that the main effect is diminished or offset by the interaction. Similarly, if you look at the interaction from the perspective of x_2 , its main effect is either enhanced or offset by the presence of x_1 in the interaction term.

B x C Interactions

Essentially, the interaction term changes the slope of the main effect. When both variables are continuous, the total effect of x_1 changes for every value of x_2 , which makes **Continuous x Continuous (C x C)** interactions difficult to interpret and require additional modeling transformations. I provide some guidance of how to build C x C interactions below and in the main book chapter for this appendix.

In this section, I focus on **B x C** interactions for two main reasons. First, **B x C** interactions are very popular, widely used and very useful to help understand effect variations across categories. Second, understanding how to specify and interpret B x C interactions is a necessary precondition to understand **C x C** interactions. **B x C** interactions have a simple and intuitive interpretation because the interaction term measures how much the slope of the main effect of the C variable changes when the B variable changes from 0 to 1. Think of it as 2 regression lines, just like with the dummy variable regression model example discussed in Chapter 3, except that the regression lines are no longer parallel, and it is the interaction term that causes the slope or effect of the regression line to change.

Let's first load the **{ISLR}** library and do some setup and data transformations with the **Auto** data set. We will use this data set to predict the gas mileage performance in vehicles

```
library(ISLR) # Contains the Auto data set
options(scipen = 4)
```

The **Auto** data set has an **origin** variable that takes the value of 1 for American vehicles and 2 and 3 for European and Japanese respectively. To illustrate the interaction model, I will convert this variable into a **foreign** variable with the value of 0 for American vehicles and 1 for all other vehicles.

```
Auto$foreign <- ifelse(Auto$origin == 1, 0, 1)
```

Let's start with a plain **linear model** without interaction terms.

```
fit.linear <- lm(mpg ~ foreign + weight + year + horsepower,
                 data = Auto)
summary(fit.linear)
```

```
Call:
lm(formula = mpg ~ foreign + weight + year + horsepower, data = Auto)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-9.4008	-2.1335	-0.0628	1.8284	13.6148

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-17.5775640	4.1432258	-4.242	0.00002769 ***
foreign	2.1155598	0.4369972	4.841	0.00000187 ***
weight	-0.0056080	0.0004338	-12.928	< 2e-16 ***
year	0.7596272	0.0507256	14.975	< 2e-16 ***
horsepower	-0.0075412	0.0091924	-0.820	0.413

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.335 on 387 degrees of freedom

Multiple R-squared: 0.8193, Adjusted R-squared: 0.8174

F-statistic: 438.6 on 4 and 387 DF, p-value: < 2.2e-16

The regression output indicates that, other things being equal, each additional pound of vehicle weight is associated a decrease of 0.0056 miles per gallon. Also, on average, foreign vehicles are more fuel efficient, yielding 2.115 more miles per gallon than domestic vehicles.

Let's add one B x C interaction term **foreign (B) * weight (C)** to evaluate if the effect of weight varies by vehicle origin.

```
fit.inter.1 <- lm(mpg ~ foreign + weight + year + horsepower +
                  foreign * weight,
                  data = Auto)
summary(fit.inter.1)
```

Call:

```
lm(formula = mpg ~ foreign + weight + year + horsepower + foreign *
    weight, data = Auto)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-9.7988	-2.0742	-0.1083	1.6348	12.8584

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-21.7924516	4.1018525	-5.313	1.83e-07 ***
foreign	10.9062409	1.7838010	6.114	2.38e-09 ***
weight	-0.0048765	0.0004446	-10.969	< 2e-16 ***
year	0.7896218	0.0495323	15.942	< 2e-16 ***
horsepower	-0.0119039	0.0089534	-1.330	0.184
foreign:weight	-0.0035444	0.0006987	-5.073	6.09e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.234 on 386 degrees of freedom

Multiple R-squared: 0.8306, Adjusted R-squared: 0.8284

F-statistic: 378.4 on 5 and 386 DF, p-value: < 2.2e-16

R is smart and knows that interaction effects should be interpreted jointly with the main effects, so if you omit the main effects of **weight** and **foreign**, R will include them because they are in the interaction term, which simplifies the model specification. This model will yield the same results as above.

```
fit.inter.1 <- lm(mpg ~ horsepower + year +
                  foreign * weight,
                  data = Auto)
summary(fit.inter.1)
```

Call:

```
lm(formula = mpg ~ horsepower + year + foreign * weight, data = Auto)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-9.7988	-2.0742	-0.1083	1.6348	12.8584

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-21.7924516	4.1018525	-5.313	1.83e-07 ***
horsepower	-0.0119039	0.0089534	-1.330	0.184
year	0.7896218	0.0495323	15.942	< 2e-16 ***
foreign	10.9062409	1.7838010	6.114	2.38e-09 ***
weight	-0.0048765	0.0004446	-10.969	< 2e-16 ***
foreign:weight	-0.0035444	0.0006987	-5.073	6.09e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.234 on 386 degrees of freedom

Multiple R-squared: 0.8306, Adjusted R-squared: 0.8284

F-statistic: 378.4 on 5 and 386 DF, p-value: < 2.2e-16

In the output above the item **foreign:weight** is the interaction term. The output shows that, on average, for each additional pound of weight, the miles per gallon performance decreases by 0.0049. The **foreign:weight** coefficient has the same sign, indicating that the effect of weight is stronger for foreign vehicles than domestic vehicles, by 0.0035. In other words, the negative effect of weight is stronger for foreign vehicles ($-0.0049 - 0.0035 = -0.0084$) than for domestic vehicles (-0.0049). Naturally, you can interpret the interaction effect from the perspective of the **foreign** predictor. In simple terms, foreign vehicles have a higher gas mileage, but this effect is offset as vehicles get heavier.

Technical Note: if for some strange reason you want to exclude the main effects and include just the interaction effect, you can use **foreign : weight** instead of **foreign * weight**. Notice the absence of main effect for foreign and weight.

```
fit.inter.noGood <- lm(mpg ~ horsepower + year +
                       foreign : weight,
                       data = Auto)
summary(fit.inter.noGood)
```

Call:

```
lm(formula = mpg ~ horsepower + year + foreign:weight, data = Auto)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-10.2516	-2.7776	-0.3643	2.5997	14.2558

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-17.502824	5.108117	-3.426	0.000677 ***
horsepower	-0.112976	0.006593	-17.136	< 2e-16 ***
year	0.678486	0.062762	10.810	< 2e-16 ***
foreign:weight	0.001380	0.000202	6.831	3.28e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.151 on 388 degrees of freedom

Multiple R-squared: 0.7193, Adjusted R-squared: 0.7171

F-statistic: 331.4 on 3 and 388 DF, p-value: < 2.2e-16

We can include more than one interaction term in the regression. For example, let's add another B x C interaction term **foreign x year**

```
fit.inter.2 <- lm(mpg ~ foreign + weight + year + horsepower +
                  foreign * weight +
                  foreign * year,
                  data = Auto)
summary(fit.inter.2)
```

Call:

```
lm(formula = mpg ~ foreign + weight + year + horsepower + foreign *
    weight + foreign * year, data = Auto)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.8995	-1.9335	-0.0871	1.5693	12.0042

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.2627664	5.1818018	-1.402	0.16184
foreign	-21.6332954	7.5624162	-2.861	0.00446 **
weight	-0.0050669	0.0004364	-11.611	< 2e-16 ***
year	0.6112655	0.0629915	9.704	< 2e-16 ***
horsepower	-0.0153070	0.0087795	-1.743	0.08205 .
foreign:weight	-0.0033460	0.0006839	-4.892	0.00000147 ***
foreign:year	0.4167118	0.0942411	4.422	0.00001275 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.158 on 385 degrees of freedom

Multiple R-squared: 0.8388, Adjusted R-squared: 0.8362

F-statistic: 333.8 on 6 and 385 DF, p-value: < 2.2e-16

The effect of **weight** is still negative, but now the effect of **foreign** is negative and quite large. Let's analyze what is happening. The interaction effect of **weight x foreign** is still negative, so the negative effect of weight is stronger for foreign vehicles, like in the model above. On average, foreign vehicles have lower gas mileage performance and this effect is enhanced for heavier vehicles.

Now, let's analyze the other interaction term. As one would expect, newer vehicles have better gas mileage, that is 0.6113 more miles per gallon for each model year. This effect is enhanced by 0.4167 for foreign vehicles. That is newer vehicle models are more fuel efficient and more so for foreign vehicles. Looking at it from the perspective of vehicle origin, foreign vehicles have a lower gas mileage, but not so much with newer vehicles.

Now let's do some ANOVA testing to see which model has more predictive power.

```
anova(fit.linear, fit.inter.1, fit.inter.2)
```

Analysis of Variance Table

Model 1: mpg ~ foreign + weight + year + horsepower

Model 2: mpg ~ horsepower + year + foreign * weight

Model 3: mpg ~ foreign + weight + year + horsepower + foreign * weight +
foreign * year

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	387	4304.9				
2	386	4035.8	1	269.10	26.975	0.0000003348 ***
3	385	3840.8	1	195.05	19.552	0.0000127533 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

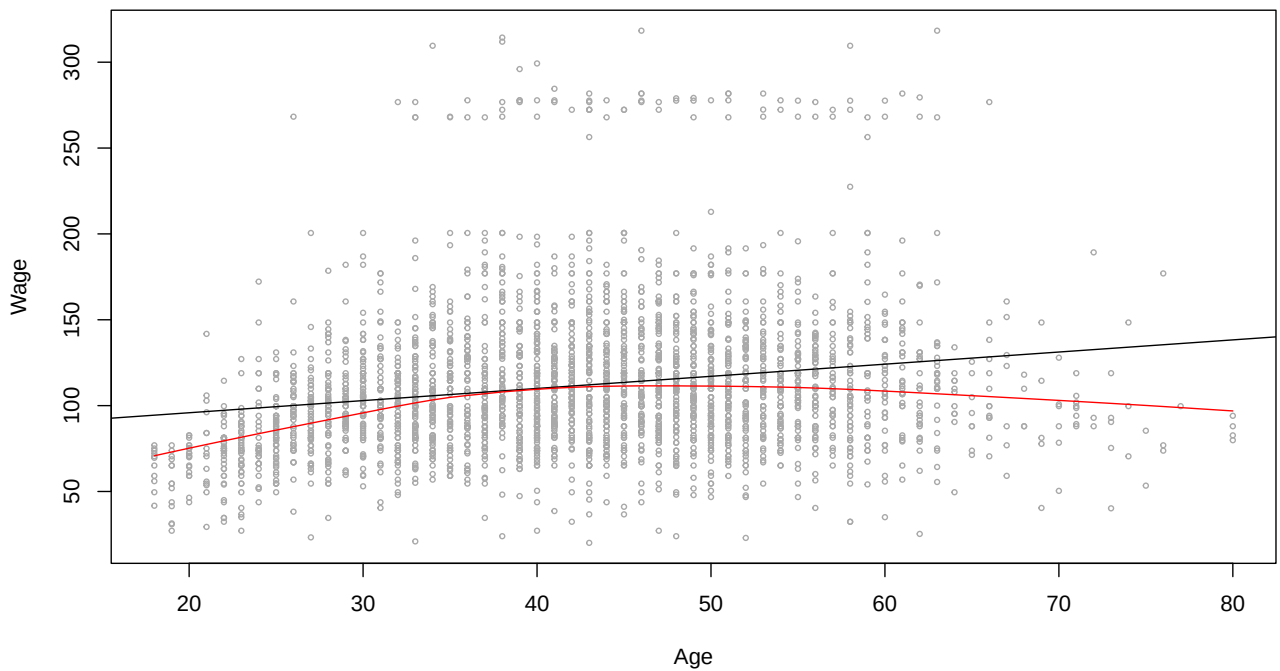
Both ANOVA p-values are highly significant, so the model with one interaction term is superior to the linear model, but the model with two interaction terms is superior to the model with one interaction term, so it is even better.

Polynomial Regression

A first step before considering a polynomial model is to plot the data. In this case, we are using the **Wage** data set from the **{ISLR}** library. I first render a bare scatter plot and I then use the **lines()** and **lowess()** functions to draw a trend line. The **lowess()** function uses **locally weighted smoothing**, which is like dividing the data into small segments and fitting mini-regression lines in each, connecting them into a “smooth” line on the plot:

```
library(ISLR)
lm.fit <- lm(wage ~ age, data = Wage) # Linear Model

plot(Wage$age, Wage$wage,
     xlab = "Age",
     ylab = "Wage",
     cex = 0.5,
     col = "darkgrey")
abline(lm.fit) # Straight line in black
lines(lowess(Wage$age, Wage$wage),
     col = "red") # Trend curve in red
```



You can change the smoothing on the line with the parameter `f`, which is the proportion of data points in each segment. The default is `f = 2/3`. Try a very small value (e.g., 0.01) for little smoothing (i.e., a jagged line) and a large one (e.g. 0.90) for a lot of smoothing (I give a straight line).

A casual inspection of the plot suggests that the relationship between wage and age is not linear. Let's start simple, with a linear model:

```
fit.poly.1 <- lm(wage ~ age,
                 data = Wage)
summary(fit.poly.1)
```

Call:

```
lm(formula = wage ~ age, data = Wage)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-100.265	-25.115	-6.063	16.601	205.748

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	81.70474	2.84624	28.71	<2e-16 ***
age	0.70728	0.06475	10.92	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.93 on 2998 degrees of freedom

Multiple R-squared: 0.03827, Adjusted R-squared: 0.03795

F-statistic: 119.3 on 1 and 2998 DF, p-value: < 2.2e-16

The linear model has a good fit, although with a very low R-Square. The output shows that wage increases with age. For each additional year of age, the wage goes up by \$707 (wages are in thousands).

Let's try a quadratic model

```
fit.poly.2 <- lm(wage ~ poly(age, 2, raw = T),
                 data = Wage)
summary(fit.poly.2)
```

Call:

```
lm(formula = wage ~ poly(age, 2, raw = T), data = Wage)
```

Residuals:

Min	1Q	Median	3Q	Max
-99.126	-24.309	-5.017	15.494	205.621

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10.425224	8.189780	-1.273	0.203
poly(age, 2, raw = T)1	5.294030	0.388689	13.620	<2e-16 ***
poly(age, 2, raw = T)2	-0.053005	0.004432	-11.960	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.99 on 2997 degrees of freedom

Multiple R-squared: 0.08209, Adjusted R-squared: 0.08147

F-statistic: 134 on 2 and 2997 DF, p-value: < 2.2e-16

This model also has a good fit with a highly significant p-value for the full model. The R-Squared increased a small amount. Notice that I used the attribute `raw = T` to obtain the raw, rather than the orthogonal coefficients. The linear effect is positive and significant, and the quadratic term is negative and also highly significant. This means that salaries start going up at the rate of \$5,294 for each additional year of age, but the negative effect of age squared makes this rate diminishing as age goes up, reaching a peak at $-5,294 / (2 * (-0.053)) = 49.9$ or almost 50 years of age, then diminishing after that.

Let's fit a cubic polynomial

```
fit.poly.3 <- lm(wage ~ poly(age, 3, raw = T),
                 data = Wage)
summary(fit.poly.3)
```

Call:

```
lm(formula = wage ~ poly(age, 3, raw = T), data = Wage)
```

Residuals:

Min	1Q	Median	3Q	Max
-99.693	-24.562	-5.222	15.096	206.119

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-75.2439142	22.1837300	-3.392	0.000703 ***
poly(age, 3, raw = T)1	10.1899915	1.6052282	6.348	2.51e-10 ***
poly(age, 3, raw = T)2	-0.1680286	0.0368602	-4.559	5.36e-06 ***
poly(age, 3, raw = T)3	0.0008495	0.0002702	3.143	0.001687 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.93 on 2996 degrees of freedom
Multiple R-squared: 0.0851, Adjusted R-squared: 0.08419
F-statistic: 92.89 on 3 and 2996 DF, p-value: < 2.2e-16

The linear and quadratic effects in the cubic polynomial model are similar than those of the quadratic model, except that the linear and quadratic effects are larger. Still, the cubic term is also positive and significant, indicating that when the wage goes up, the cubic effect starts pulling the curve upwards. However, the cubic effect is too small to observe a valley in the present data range.

Let's evaluate the 3 models with an ANOVA test

```
anova(fit.poly.1, fit.poly.2, fit.poly.3)
```

Analysis of Variance Table

```
Model 1: wage ~ age
Model 2: wage ~ poly(age, 2, raw = T)
Model 3: wage ~ poly(age, 3, raw = T)
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1    2998 5022216
2    2997 4793430   1    228786 143.4679 < 2.2e-16 ***
3    2996 4777674   1     15756   9.8801 0.001687 **
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The first ANOVA test is highly significant, indicating that the quadratic model has much higher predictive power. The second ANOVA test is also significant, but not as much. Nevertheless, the cubic model is still better than the quadratic model because it has the highest predictive power, significantly higher than the quadratic model. You can keep trying higher polynomials and testing them with ANOVA, but a better test would be to evaluate the models with the CV test error.

A note about **raw polynomials**: by default, **raw = F**, which causes the `poly()` to create **orthogonal polynomial** terms. These orthogonal polynomials are simply the principal components of the raw polynomial variables, which is meant to help reduce multi-collinearity. Setting **raw = T** computes back the raw coefficients, which are what we need for interpretation.

Alternatively, you could fit the same model above the long way with the `I()` (identity) function and get identical results, but with a lot more hand coding:

```
fit.poly.3I <- lm(wage ~ age + I(age ^ 2) + I(age ^ 3),
                  data = Wage)
summary(fit.poly.3I)
```

Call:

```
lm(formula = wage ~ age + I(age^2) + I(age^3), data = Wage)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-99.693	-24.562	-5.222	15.096	206.119

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-75.2439142	22.1837300	-3.392	0.000703 ***
age	10.1899915	1.6052282	6.348	2.51e-10 ***

```

I(age^2)      -0.1680286    0.0368602   -4.559 5.36e-06 ***
I(age^3)      0.0008495    0.0002702    3.143 0.001687 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.93 on 2996 degrees of freedom
Multiple R-squared:  0.0851,    Adjusted R-squared:  0.08419
F-statistic: 92.89 on 3 and 2996 DF,  p-value: < 2.2e-16

```

Predictions and CV Testing with Polynomials

You can use the `cv.glm()` function to compute LOOCV, KFCV or any other CV test errors. In this section, I will use random subset sampling to do quick tests and compare the three polynomials, and also to illustrate how to do predictions.

```

RNGkind(sample.kind = "default") # R default RNG
set.seed(1) # Set any random seed

```

Let's construct a 80% train and 20% test subsets and make predictions

```

library(ISLR)
tr.size <- 0.80

train <- sample(nrow(Wage),
               tr.size * nrow(Wage))

Wage.train <- Wage[train, ] # Train subset
nrow(Wage.train) # 2,400 observations

```

```
[1] 2400
```

```

Wage.test <- Wage[-train, ] # Test subset
nrow(Wage.test) # 600 observations

```

```
[1] 600
```

Now let's train, predict and test the 3 polynomial models

```

# Train the models

fit.poly.1.train <- lm(wage ~ age,
                      data = Wage.train)
fit.poly.2.train <- lm(wage ~ poly(age, 2, raw = T),
                      data = Wage.train)
fit.poly.3.train <- lm(wage ~ poly(age, 3, raw = T),
                      data = Wage.train)

```

Now let's predict with test data with each of the 3 trained models

```

wage.pred.1 <- predict(fit.poly.1.train, Wage.test)
wage.pred.2 <- predict(fit.poly.2.train, Wage.test)
wage.pred.3 <- predict(fit.poly.3.train, Wage.test)

```

Then compute the MSE for each and display the results

```
mse.test.1 <- mean( (Wage.test$wage - wage.pred.1) ^ 2)
mse.test.2 <- mean( (Wage.test$wage - wage.pred.2) ^ 2)
mse.test.3 <- mean( (Wage.test$wage - wage.pred.3) ^ 2)
cbind("CV MSE Linear" = mse.test.1,
      "CV MSE Quadratic" = mse.test.2,
      "CV MSE Cubic" = mse.test.3)
```

```
      CV MSE Linear CV MSE Quadratic CV MSE Cubic
[1,]      2068.485      1989.221      1986.052
```

The CV test MSE yield very similar results to the ANOVA tests we did earlier. The quadratic model is superior to the linear model, and the cubic model is slightly better than the quadratic model.

Step Functions

A **Step** function fits a bunch of horizontal lines in different **segments** of the data. The points where the partitions begin and end are the **knots**. To fit a Step function, we can use the **cut()** function to arbitrarily divide the age data into, say 4 partitions.

Technical Note: A **Step function** is a horizontal linear model that changes vertical positions at various points, just like steps in a ladder. This is completely different than and should not be confused with the **Stepwise** method, which is not a model, but a variable selection method.

```
library(ISLR)
fit.step <- lm(wage ~ cut(age, 4),
              data = Wage)
summary(fit.step) # Take a look (coefficients are relative to intercept)
```

Call:

```
lm(formula = wage ~ cut(age, 4), data = Wage)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-98.126	-24.803	-6.177	16.493	200.519

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	94.158	1.476	63.790	<2e-16 ***
cut(age, 4)(33.5,49]	24.053	1.829	13.148	<2e-16 ***
cut(age, 4)(49,64.5]	23.665	2.068	11.443	<2e-16 ***
cut(age, 4)(64.5,80.1]	7.641	4.987	1.532	0.126

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.42 on 2996 degrees of freedom

Multiple R-squared: 0.0625, Adjusted R-squared: 0.06156

F-statistic: 66.58 on 3 and 2996 DF, p-value: < 2.2e-16

Let's do predictions and CV testing on this model

```
fit.step.train <- lm(wage ~ cut(age, 4),
                    data = Wage.train)
pred.step <- predict(fit.step.train, Wage.test)
mse.step <- mean( (Wage.test$wage - pred.step) ^ 2)
mse.step # Take a look
```

[1] 2021.074

The Step function has more predictive accuracy than the linear model, but less accuracy than the quadratic or the cubic models. Makes sense.

Piecewise Linear Spline Regression

There are a few functions in R to fit piecewise linear and spline regression models. In this illustration, I code the piecewise linear model from scratch to describe how this works more effectively. Also, technically, a piecewise model is a regression divided into segments, with a different slope in each of the segments. A Spline model is similar to piecewise, but the model imposes the constraint that lines to meet at the knot (i.e., where two segments meet). If we don't impose this constraints, the model will have two different predictions at each knot, one for each line meeting at the know. This is why splines are preferred. In this Appendix, I use the terms piecewise and spline interchangeable and assume the constraint that regression line segments must meet at the knots.

Let's use the same segments discussed in the main chapter at ages 25, 40 and 60. Note that I use the `I()` function. This is the **Identity** function, which basically doesn't do anything and doesn't change anything, but it is useful because sometimes the operations inside the functions can have other meaning in R. The identity function simply says "do this". The `I()` function creates the respective dummy variables and interaction terms. Let's discuss each segment of the piecewise linear model:

- First Segment: we simply model **age** as a predictor. This will give us a slope for the regression line starting at the intercept. However, we will add an interaction term to change this slope in the next segment after **age** = 25.
- Second Segment: we include an interaction term $(\text{age} - 25) * (\text{age} > 25)$ (wrapped inside the identity `I()` function). If age is ≤ 25 , then $(\text{age} > 25)$ is 0, so there is no effect of this term in the first segment, and it is 1 after age 25. The term $(\text{age} - 25)$ simply moves the Y axis to the knot at 25. In sum $(\text{age} - 25) * (\text{age} > 25)$ changes the slope of the line at **age** = 25 and does nothing before that.
- Third and Fourth Segments: do the same thing, but at the knots where **age** = 40 and **age** = 60 respectively.

Technical Note: the coefficients of the respective segments below **do not** represent the effect in that segment, but the **change in effect** after the knot. To find the total effect in the second segment you need to add the effect coefficients of the first and second segment. For the total effect in the third segments is equal to the sum of the coefficients for the first three segments, and so on.

```
library(ISLR)
fit.piecewise <- lm(wage ~ age +
                   I((age - 25) * (age > 25)) +
                   I((age - 40) * (age > 40)) +
                   I((age - 60) * (age > 60)),
                   data = Wage)
summary(fit.piecewise) # Take a look
```

Call:

```
lm(formula = wage ~ age + I((age - 25) * (age > 25)) + I((age -
  40) * (age > 40)) + I((age - 60) * (age > 60)), data = Wage)
```

Residuals:

Min	1Q	Median	3Q	Max
-99.795	-24.686	-4.856	15.344	204.671

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-42.4688	23.2695	-1.825	0.06809 .
age	5.3779	0.9739	5.522	3.64e-08 ***
I((age - 25) * (age > 25))	-3.4977	1.0800	-3.239	0.00121 **
I((age - 40) * (age > 40))	-1.9800	0.2970	-6.667	3.10e-11 ***
I((age - 60) * (age > 60))	-1.4041	0.5424	-2.589	0.00968 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.91 on 2995 degrees of freedom

Multiple R-squared: 0.08665, Adjusted R-squared: 0.08543

F-statistic: 71.03 on 4 and 2995 DF, p-value: < 2.2e-16

The results indicate that the linear regression line starts with \$5.38K more wage dollars for every additional year of age, up to age **25**. At that point the slope changes to $5.38 - 3.50 = 1.88$ K higher wage per year of age, up to age **40**. It then changes to $1.88 - 1.98 = -0.10$ K lower wage for every additional year of age (i.e., almost flat), up to age **60**. After that, the wage declines at the rate of $-0.10 - 1.40 = -1.50$ K fewer dollars for each additional year of age.

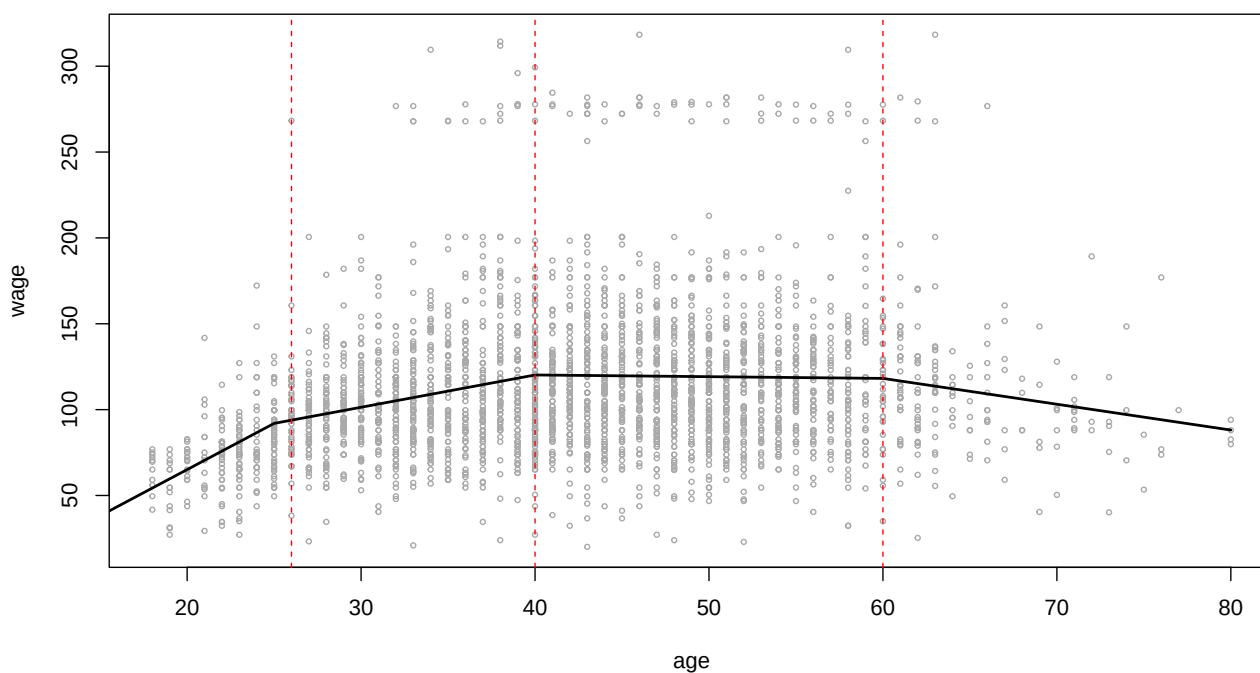
The procedure for any distribution of knots and segments is the same. That is, include the predictor (e.g., age) first and then add an interaction term at each knot where you want the slope of the regression model to change.

Visualizing the Linear Spline Model

Piecewise linear splines are useful to visualize. To do this, create a vector with age values of interest. Because the data range is from 0 to 80, we include these age points -> 0, 25, 40, 60 and 80. Let's then predict wages at the age knots and use the results to plot a graph and draw lines connecting the age knots with vertical dashed lines at each knot:

```
age.knots <- c(0, 25, 40, 60, 80)
preds <- predict(fit.piecewise,
                 list(age = age.knots),
                 se = T) # se = T computes standard errors

plot(Wage$age, Wage$wage,
     cex = 0.5,
     col = "darkgrey",
     xlab = "age",
     ylab = "wage")
lines(age.knots, preds$fit, lwd = 2)
abline(v = 26, col = "red", lty = "dashed") # At age 26
abline(v = 40, col = "red", lty = "dashed") # At age 40
abline(v = 60, col = "red", lty = "dashed") # At age 60
```



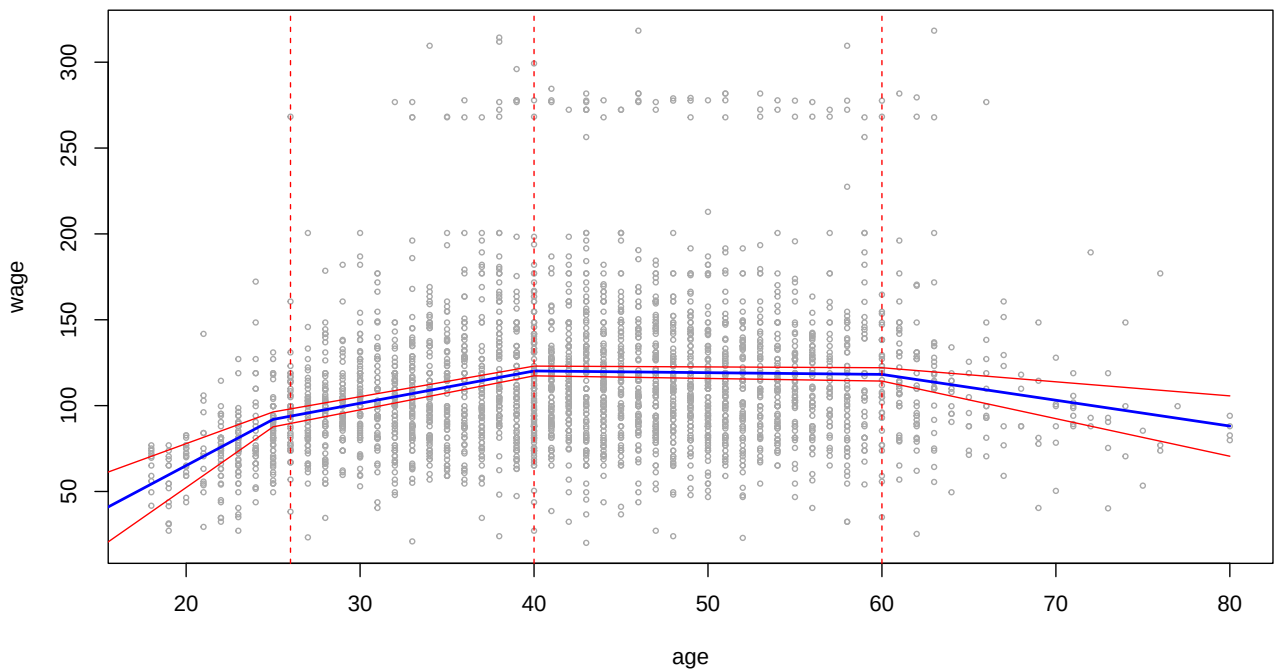
Now let's plot the same graph, but with 95% confidence interval bands using the standard errors in the `se` parameter, that is $\pm 2 * \text{preds\$se.fit}$.

```
plot(Wage$age, Wage$wage,
     cex = 0.5,
     col = "darkgrey",
     xlab = "age",
     ylab = "wage")

lines(age.knots, preds$fit,
      col = "blue",
      lwd = 2)

abline(v = 26, col = "red", lty = "dashed") # At age 26
abline(v = 40, col = "red", lty = "dashed") # At age 40
abline(v = 60, col = "red", lty = "dashed") # At age 60

lines(age.knots,
      preds$fit - 2 * preds$se.fit,
      col = "red",
      lwd = 1)
lines(age.knots,
      preds$fit + 2 * preds$se.fit,
      col = "red", lwd = 1)
```



CV Error Testing

```
fit.piecewise.train <- lm(wage ~ age +
                          I((age - 25) * (age > 25)) +
                          I((age - 40) * (age > 40)) +
                          I((age - 60) * (age > 60)),
                          data = Wage.train)
pred.piecewise <- predict(fit.piecewise.train, Wage.test)
mse.piecewise <- mean((Wage.test$wage - pred.piecewise) ^ 2)
mse.piecewise # Take a look
```

```
[1] 1978.821
```

Let's test the model with LOOCV and 10FCV

```
library(boot) # Contains the cv.glm() function
glm.fit.piece <- glm(wage ~ age +
                    I((age - 25) * (age > 25)) +
                    I((age - 40) * (age > 40)) +
                    I((age - 60) * (age > 60)),
                    data = Wage)

cv.loo <- cv.glm(Wage, glm.fit.piece)
test.mse.loo <- round(cv.loo$delta[1], digits = 3) # Round and save it

cv.10F <- cv.glm(Wage, glm.fit.piece, K = 10)
test.mse.10F <- round(cv.10F$delta[1], digits = 3) # Round and save it
```

Let's print all CV Test MSE results together, including the quadratic model and compare

```
cbind("CV MSE Split Quadr" = mse.test.2,
      "CV MSE Split Piece" = mse.piecewise,
      "LOOCV" = test.mse.loo,
      "10FCV" = test.mse.10F)
```

```
      CV MSE Split Quadr CV MSE Split Piece      LOOCV      10FCV
[1,]          1989.221          1978.821 1593.894 1595.176
```

Interestingly, the piecewise linear spline is more accurate than the quadratic model, but a little less accurate than the cubic model. But the piecewise linear spline is much more interpretable than the cubic model.

Multivariate Adaptive Regression Spline (MARS)

The piecewise linear spline model above is very interpretable, and tend to has better fit than polynomials and step functions because we can set as many knots as we want and therefore control the slopes in the various resulting segments. But if we are not so interested in interpretation, and are more interested in making predictions, MARS **B-Spline** models are a good alternative option. MARS models are easier to formulate and we can not only fit linear splines, but also polynomial splines connecting knots with curves, rather than with straight lines. I will use the **basis spline** function `bs()` from the `{splines}` package for this purpose. You can use the `summary()` function to view the regression output, if you wish, but the interpretation is not very intuitive.

Linear B-Spline

Let's train the model, make predictions and do CV testing

```
library(ISLR)
library(splines) # Needed to fit spline models
fit.bs1.train <- lm(wage ~ bs(age,
                             knots = c(25, 40, 60),
                             degree = 1), # Linear B-Spline (degree = 1)
                   data = Wage.train)
pred.bs1 <- predict(fit.bs1.train, Wage.test)
mse.bs1 <- mean( (Wage.test$wage - pred.bs1) ^ 2)
mse.bs1 # Take a look
```

```
[1] 1978.821
```

It's like magic !! Exact same result as the piecewise linear spline above

Polynomial and Cubic (Basis) B-Splines

Let's now try a **Cubic B-Spline**. You can try other polynomials by changing the `degree =` parameter. Notice that I don't use the `degree = 3` parameter for the cubic spline because it is the default for the `bs()` and `ns()` functions.

```
fit.bs3.train <- lm(wage ~ bs(age,
                             knots = c(25, 40, 60)),
                   data = Wage.train)
pred.bs3 <- predict(fit.bs3.train, Wage.test)
mse.bs3 <- mean( (Wage.test$wage - pred.bs3) ^ 2)
mse.bs3 # Take a look
```



```
[1] 1991.037
```

A tad more accurate than the linear spline model

Polynomial and Cubic (Natural) N-Splines

N-Splines are identical to B-Splines, except for the first and last segments, but they tend to give very similar predictions. The main difference is that `bs()` fits polynomial functions of the degree specified in all segments, whereas `ns()` does the same, but fits linear models in the first and last segments to reduce the “tail-wagging” effects in the confidence intervals (i.e., errors tend to be larger at both tail ends in curvilinear models).

Let’s fit a linear and a cubic N-Spline, and do predictions and CV testing. The only thing that we need to do differently is to use the natural spline function `ns()` instead of the basis spline function `bs()`. Note: the linear B-Spline and linear N-Spline are identical, so I don’t model the linear N-Spline below, only the cubic.

Cubic N-Spline

```
# Cubic N-Spline (default is degree = 3)
fit.ns3.train <- lm(wage ~ ns(age,
                             knots = c(25, 40, 60)),
                   data = Wage.train)
pred.ns3 <- predict(fit.ns3.train, Wage.test)
mse.ns3 <- mean( (Wage.test$wage - pred.ns3) ^ 2)
```

Print ALL MSEs

```
cbind("CV MSE Linear" = mse.test.1,
      "Poly(2)" = mse.test.2,
      "Poly(3)" = mse.test.3,
      "Step" = mse.step,
      "Piecewise" = mse.piecewise)
```

```
      CV MSE Linear Poly(2) Poly(3)      Step Piecewise
[1,]      2068.485 1989.221 1986.052 2021.074  1978.821
```

```
cat("\n")
```

```
cbind("Linear B-Spline" = mse.bs1,
      "Cubic B-Spline" = mse.bs3,
      "Cubic N-Spline" = mse.ns3)
```

```
      Linear B-Spline Cubic B-Spline Cubic N-Spline
[1,]      1978.821      1991.037      1986.576
```

Of all the models, the **Cubic B-Spline** is the most accurate in this example.